Mid-term Exam Review

Science & Technology Innovation Management

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Exam: 23/10/2026, 14:55–15:55 (written, 1 hour)

1. Definition of innovation

"Definition of innovation."

EN: Innovation is the successful exploitation of new ideas. It is the full journey from an idea to real, delivered value, not just the moment a new idea appears. Three terms are easy to confuse and worth separating clearly. **Invention** is the creation of something genuinely new, a new idea, device, or technology, often born in a lab or a research project. **Innovation** is what happens next: taking that invention and turning it into something practical that customers will actually buy, or that solves a real social problem, so it generates commercial or social value. An invention that never reaches a market or a user is not yet an innovation. **Diffusion** is the stage after that: the gradual process by which the wider market learns about the innovation, trusts it, and adopts it, spreading from a small group of early users to the mass market over time. Innovation management, as a discipline, is concerned mainly with the middle step: deliberately organizing people, resources, and decisions so that promising ideas survive the journey from invention to adopted, value-creating innovation.

VN: Đổi mới sáng tạo (innovation) là việc khai thác thành công một ý tưởng mới, tức là toàn bộ hành trình từ ý tưởng đến giá trị thực sự được tạo ra, chứ không chỉ là khoảnh khắc nảy ra ý tưởng đó. Ba khái niệm dễ nhầm lẫn và cần phân biệt rõ. Sáng chế (invention) là việc tạo ra một thứ hoàn toàn mới, một ý tưởng, thiết bị hay công nghệ mới, thường ra đời trong phòng thí nghiệm hoặc một dự án nghiên cứu. Đổi mới sáng tạo (innovation) là bước tiếp theo: biến sáng chế đó thành thứ có ích thực tế mà khách hàng sẵn sàng mua, hoặc giải quyết được một vấn đề xã hội thật, từ đó tạo ra giá trị thương mại hoặc xã hội. Một sáng chế chưa bao giờ đến được với thị trường hay người dùng thì chưa phải là một đổi mới sáng tạo. Khuếch tán (diffusion) là giai đoạn sau đó: quá trình thị trường dần biết đến, tin tưởng và chấp nhận đổi mới sáng tạo đó, lan rộng từ một nhóm nhỏ người dùng đầu tiên ra đến đại chúng theo thời gian. Quản trị đổi mới sáng tạo, với tư cách một lĩnh vực, chủ yếu quan tâm đến bước ở giữa: tổ chức con người, nguồn lực và các quyết định một cách có chủ đích để những ý tưởng tiềm năng có thể sống sót qua hành trình từ sáng chế đến một đổi mới sáng tạo được thị trường chấp nhận và tạo ra giá trị.

2. The "4Ps" model (Tidd & Bessant)

"What does the "4Ps of Innovation" model by Tidd & Bessant entail? Provide an example of how the innovation strategies of Product, Process, Position, and Paradigm are applied at a major technology company (such as Apple, Viettel, or FPT)."

Tidd and Bessant argue that innovation is not limited to inventing a new gadget; a company can innovate along four different dimensions, often at the same time.

Product innovation changes what the organization actually offers to its customers, a new or significantly improved good or service. *Đổi mới sản phẩm thay đổi những gì tổ chức cung cấp cho khách hàng.* **Process** innovation changes how that product or service is created and delivered, for example a more efficient, cheaper, or higher-quality way of manufacturing or operating. *Đổi mới quy trình thay đổi cách sản phẩm hay dịch vụ được tạo ra và phân phối.* **Position** innovation changes the context in which the product is introduced, who it is sold to, which market segment, which geography, or how it is positioned against competitors, even if the product itself barely changes. *Đổi mới vị thế thay đổi bối cảnh đưa sản phẩm ra thị trường: bán cho ai, ở đâu, định vị thế nào so với đối thủ.* **Paradigm** innovation is the deepest and rarest kind: a change in the underlying business model or the mental model of what the organization fundamentally is and how it makes money, not just what it sells. *Đổi mới mô hình (paradigm) là thay đổi sâu nhất: thay đổi chính mô hình kinh doanh hay cách tổ chức tự định nghĩa mình.*

Apple illustrates all four. Its Product innovation is visible in devices like the iPhone and in chips it designs itself rather than buying off the shelf. Its Process innovation lies in how tightly it integrates hardware, software, and chip design under one roof, instead of outsourcing each layer separately. Its Position innovation is a deliberate premium strategy: a closed, tightly controlled ecosystem aimed at a higher-paying segment rather than competing on price. Its Paradigm innovation is the clearest of the four: a gradual shift away from a company that earns money mainly from one-time hardware purchases, toward one that earns a growing share of its revenue from recurring digital services that keep customers inside its ecosystem year after year.

Viettel shows the same logic in a different industry. Its Product innovation is building its own telecom network equipment rather than only buying foreign-made gear, developed jointly with international technology partners and tested with operators abroad. Its Process innovation comes from doing this research and development in-house through its own technology arm, instead of relying purely on licensing. Its Position innovation is moving from a company that only serves the domestic Vietnamese market to one that sells its own equipment abroad, competing directly with established global telecom-equipment makers. Its Paradigm innovation is the broadest shift: evolving from being simply a telecom operator that sells calls and data, into a technology and defense-oriented conglomerate with its own hardware and research capability.

FPT shows the pattern again in the software and AI sector. Its Product innovation is building its own AI computing infrastructure and chip-related offerings, instead of only providing contracted software labor. Its Process innovation is developing and operating this AI infrastructure internally. Its Position innovation is pushing into a demanding international market, notably Japan, positioning itself as a serious AI and digital-transformation partner rather than a low-cost outsourcing vendor. Its

Paradigm innovation is the shift in identity this implies: moving away from being thought of as an outsourcing company that executes someone else's specifications, toward being seen as a genuine technology and AI partner with its own products.

3. Risks in the "Implementing" phase

"During the "Implementing" phase, what risks typically arise when transforming a technical concept into a commercial product or service? How can one manage both technological and market risks?"

The Implementing phase is where an idea that has survived earlier screening is finally committed to real resources: engineering time, capital, manufacturing or operational capacity. This is also where uncertainty is still high but the cost of being wrong rises sharply, which is why this phase concentrates two distinct families of risk.

Technological risk *(rủi ro công nghệ)* is the risk that the underlying idea simply does not hold up outside a controlled setting. A solution that works as a small prototype in a lab may not scale reliably to full production volume, may turn out far more expensive to manufacture than expected, or may behave unpredictably once it meets real-world conditions that the lab never tested for.

Market risk *(rủi ro thị trường)* is a separate and equally dangerous risk: that even a technically sound product fails commercially. Customers may not actually want it in the way the team assumed, real demand may turn out to be much smaller than projected, pricing assumptions may be wrong, or a rival firm, or an entirely different substitute, may reach the same customers first and capture the opportunity.

Managing both risks well means refusing to commit all resources at once and instead learning in stages. A **stage-gate** process breaks the journey into checkpoints, where the project must prove itself before receiving the next round of funding or resources, so a weak idea can be stopped early rather than discovered to be weak only after full-scale launch. Building a **minimum viable product or prototype** lets the team test whether the core technical idea actually works at low cost, before investing in a polished, full-featured version. Running a **small pilot** with real customers, rather than assuming demand from internal opinion, tests the market-risk side directly: does a real customer actually choose to pay for or use this, under real conditions. Critically, the two tracks should run **in parallel rather than sequentially**, testing the technology and testing the market at roughly the same time, since a product that is technically perfect but commercially unwanted is just as much a failure as one that is commercially desired but technically broken.

VN: Giai đoạn Implementing là lúc một ý tưởng đã qua sàng lọc bắt đầu được đầu tư nguồn lực thật sự, nên đây là nơi tập trung hai loại rủi ro riêng biệt. Rủi ro công nghệ là khả năng ý tưởng kỹ thuật không hoạt động tốt khi ra khỏi môi trường thử nghiệm được kiểm soát: khó mở rộng quy mô, chi phí sản xuất cao hơn dự tính, hoặc hoạt động không ổn định trong điều kiện thực tế. Rủi ro thị trường là khả năng một sản phẩm kỹ thuật tốt vẫn thất bại về thương mại: khách hàng không thực sự cần nó như nhóm phát triển giả định, nhu cầu thực tế nhỏ hơn kỳ vọng, hoặc đối thủ/sản phẩm thay thế đến trước. Để quản lý cả hai, nên chia dự án thành các mốc kiểm tra (stage-gate) để dừng sớm nếu ý tưởng yếu, làm MVP/prototype để kiểm tra tính khả thi kỹ thuật với chi phí thấp, chạy thử nghiệm nhỏ với khách hàng thật để kiểm chứng nhu cầu, và quan trọng nhất là kiểm chứng cả hai loại rủi ro song song chứ không lần lượt.

4. The role of innovation management (5 official roles)

"Analyze the role of innovation management for businesses. Provide illustrative examples."

Innovation management is not simply the act of having good ideas; it is the discipline of organizing a business so that good ideas reliably turn into results. The course frames this through five roles.

1. Creating and sustaining competitive advantage. *Tạo và duy trì lợi thế cạnh tranh.* A single clever idea can be copied. The role of innovation management here is to make advantage repeatable, building an organizational habit of continuously testing, improving, and refining, so the company keeps earning an edge rather than enjoying one brief success.

2. Integrating technology, market, and organization. *Gắn kết công nghệ, thị trường và tổ chức.* A technically brilliant solution is worthless if it does not fit what the market actually wants, and a market opportunity is worthless if the organization lacks the internal capability or structure to deliver it. This role is about deliberately connecting these three sides so they reinforce each other instead of pulling in different directions.

3. Transforming ideas into real value. *Biến ý tưởng thành giá trị thật.* Many organizations are good at generating ideas and workshops full of brainstorming, but far fewer are good at the harder, less glamorous work of actually shipping something that customers pay for or benefit from. This role is the discipline of following through past the idea stage to an outcome that creates measurable value.

4. Building and optimizing dynamic capabilities. *Xây dựng và tối ưu năng lực thích ứng.* Markets, technologies, and customer expectations keep shifting. This role is about building the organizational muscle to adapt, to reorganize resources, retrain teams, or redesign processes as conditions change, rather than being optimized only for yesterday's situation.

5. Managing risk and optimizing resources. *Quản lý rủi ro và tối ưu nguồn lực.* Growth and experimentation inherently carry risk. This role is about expanding and innovating responsibly, so that the pursuit of new opportunities does not create risk exposure that threatens the core business, while still spending resources efficiently rather than wastefully.

Illustrative example – Grab. Grab is a useful single example because it shows all five roles working together inside one company. Its continuous, data-driven experimentation culture, constantly testing and refining how it matches drivers with riders and how it prices rides, is an example of sustaining competitive advantage rather than relying on a one-time breakthrough. Its super-app strategy, bundling rides, food delivery, payments, and financial services into one platform with one account and

one set of users, is a clear case of integrating technology, market reach, and organizational structure into a single reinforcing loop. Its map and location-data product, originally built only to solve an internal problem for its own drivers before later being offered as a service to other businesses, is a direct example of turning an internal idea into real external value instead of leaving it as an unused internal tool. Its gradual, disciplined reduction of promotional and incentive spending over the years, done carefully enough not to damage user growth, shows dynamic capability: adapting its spending behavior as the business matured from a growth-at-all-costs startup into a more disciplined public company. Finally, its dedicated fraud-prevention system, and the years it spent methodically obtaining financial licenses before offering lending and other financial products, show risk management done deliberately rather than carelessly, protecting the core platform while still expanding into new, riskier business lines.

Good luck this afternoon, 14:55–15:55! / Chúc may mắn chiều nay!